

# Predicting the Quality of Life Index: A Comparative XGBoost-LSTM Study with SHAP-Based Explainable AI

Replication Study (ECO613M)

Oshi Johri (251270022)

Department of Economic Sciences, IIT Kanpur

April 17, 2026

# Outline

Results: Model Performance

SHAP

Extension

# Paper Overview

**Paper:** Predicting the Quality of Life Index using XGBoost and LSTM

**Source:** Budak (2025)

## Objective

Predict Quality of Life Index using 8 sub-indices and compare:

- ▶ XGBoost (tree-based model)
- ▶ LSTM (time-series deep learning)

## Key Idea

Use SHAP to identify:

- ▶ Which features matter most

Dataset: 88 countries, 2016–2025 panel

# Dataset and Variables

## Target Variable

Quality of Life Index

## Features (8 Sub-indices)

- ▶ Purchasing Power Index
- ▶ Pollution Index
- ▶ Cost of Living Index
- ▶ Traffic Commute Time Index
- ▶ Safety Index
- ▶ Health Care Index
- ▶ Property Price to Income Ratio
- ▶ Climate Index

# Data Preprocessing

## Cleaning Steps

- ▶ Removed missing values
- ▶ Dropped irrelevant columns (Rank)

## Feature Engineering

- ▶ Reversed indices:
  - ▶ Pollution
  - ▶ Cost of Living
  - ▶ Traffic
  - ▶ Property Price Ratio

## Key Insight

Ensures all features move in same direction (higher = better QoL)

# Modeling Approach

## XGBoost

- ▶ Gradient boosted trees
- ▶ Handles non-linearity well
- ▶ No scaling needed

## LSTM

- ▶ Sequential model
- ▶ Uses past 3 years
- ▶ Captures temporal dynamics

## What is SHAP?

- ▶ Based on game theory (Shapley values)
- ▶ Measures feature contribution

---

## Results: Model Performance

---



## Descriptive Statistics (Replication)

### Summary Statistics (2020–2025)

| Variable         | Mean    | Std Dev | Min     | Max     | N  |
|------------------|---------|---------|---------|---------|----|
| Quality of Life  | 135.496 | 35.318  | 44.533  | 198.475 | 88 |
| Purchasing Power | 65.100  | 32.113  | 10.250  | 139.775 | 88 |
| Pollution        | 42.247  | 20.691  | 7.538   | 83.350  | 88 |
| Property Ratio   | 188.678 | 7.431   | 148.960 | 199.220 | 88 |
| Cost of Living   | 81.104  | 19.125  | 11.700  | 109.980 | 88 |
| Safety           | 59.727  | 13.729  | 17.320  | 85.633  | 88 |
| Health Care      | 64.153  | 10.091  | 38.580  | 85.163  | 88 |
| Traffic          | 29.305  | 8.263   | 2.500   | 44.800  | 88 |
| Climate          | 76.069  | 16.946  | 20.200  | 99.800  | 88 |

**Table:** Descriptive Statistics of Quality of Life Indices for the Period 2016–2025

| <b>Variable</b>             | <b>N</b> | <b>Mean</b> | <b>Std. Dev.</b> | <b>Min</b> | <b>Max</b> | <b>Direction</b> |
|-----------------------------|----------|-------------|------------------|------------|------------|------------------|
| Quality of Life Index       | 88       | 134.988     | 35.637           | 44.533     | 198.475    | Higher is better |
| Purchasing Power Index      | 88       | 65.177      | 32.107           | 10.250     | 139.775    | Higher is better |
| Pollution Index             | 88       | 53.768      | 20.681           | 12.650     | 88.462     | Lower is better  |
| House Price to Income Ratio | 88       | 13.420      | 7.422            | 2.880      | 53.140     | Lower is better  |
| Cost of Living Index        | 88       | 50.603      | 19.123           | 21.720     | 120.000    | Lower is better  |
| Safety Index                | 88       | 59.711      | 13.788           | 17.320     | 85.633     | Higher is better |
| Health Care Index           | 88       | 64.144      | 10.106           | 38.580     | 85.162     | Higher is better |
| Traffic Commute Time Index  | 88       | 36.024      | 8.272            | 20.500     | 62.800     | Lower is better  |
| Climate Index               | 88       | 74.944      | 18.512           | 20.200     | 99.800     | Higher is better |

# Prediction Comparison: Paper vs Replication

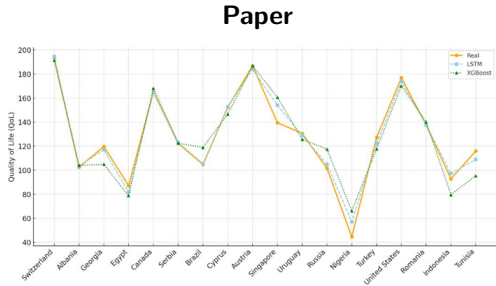
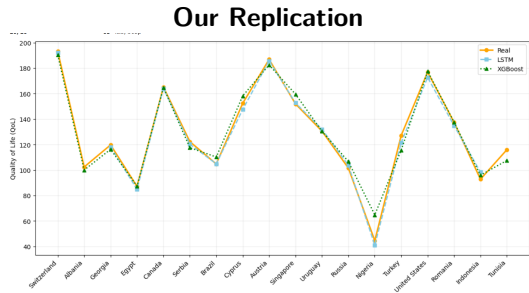


Figure 1 (Paper)



Our generated plot

# Model Performance Comparison

## Paper Results

| Model   | $R^2$ | MAE  | RMSE  | MAPE |
|---------|-------|------|-------|------|
| XGBoost | 0.898 | 9.23 | 11.73 | 9.56 |
| LSTM    | 0.978 | 3.75 | 5.44  | 4.06 |

## Our Results

| Model   | $R^2$ | MAE  | RMSE  | MAPE |
|---------|-------|------|-------|------|
| XGBoost | 0.908 | 7.65 | 10.04 | 6.28 |
| LSTM    | 0.950 | 5.92 | 7.71  | 4.54 |

## Insight

- ▶ LSTM consistently outperforms XGBoost

# Model Configuration and Training Setup

## XGBoost Hyperparameters

| Parameter     | Value |
|---------------|-------|
| n_estimators  | 50    |
| learning_rate | 0.085 |
| max_depth     | 2     |
| subsample     | 0.78  |
| reg_lambda    | 6.5   |
| random_state  | 42    |

## Deep Learning Training Setup

| Setting          | LSTM |
|------------------|------|
| Epochs           | 500  |
| Batch Size       | 8    |
| Validation Split | 0.20 |
| Early Stopping   | Yes  |
| Optimizer        | Adam |
| Loss Function    | MSE  |

---

**SHAP**

---

# SHAP Comparison: Paper vs Replication

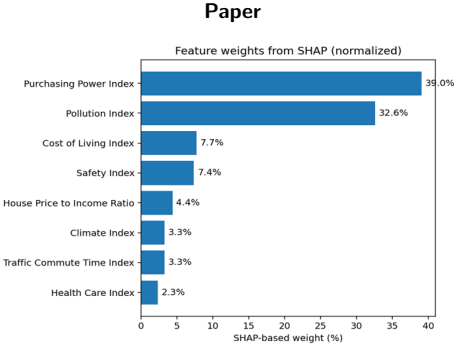
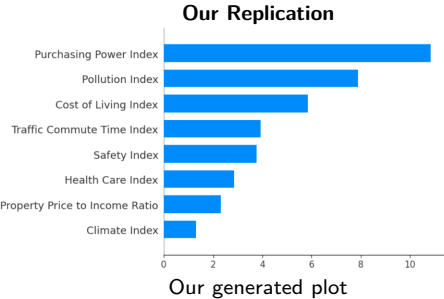


Figure 1 (Paper)



## Feature Importance Share (%)

| Feature | PP     | Pollution | Cost Living | Traffic | Safety | Health | Property | Climate |
|---------|--------|-----------|-------------|---------|--------|--------|----------|---------|
| Percent | 27.991 | 20.358    | 15.109      | 10.134  | 9.723  | 7.390  | 5.946    | 3.348   |

# SHAP Beeswarm

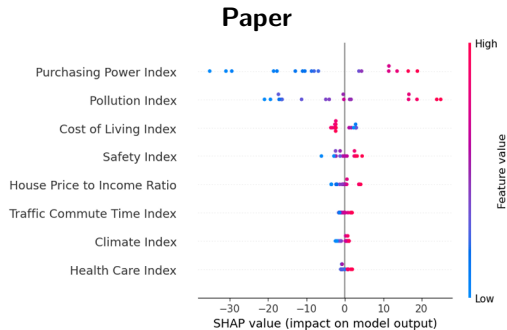
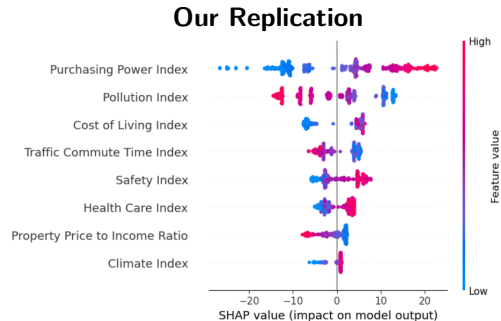


Figure 1 (Paper)



Our generated plot



---

## Extension

---

# Model Performance Comparison

Prediction Accuracy Across Models

| Model   | $R^2$  | MAE  | RMSE  | MAPE (%) |
|---------|--------|------|-------|----------|
| XGBoost | 0.9086 | 7.65 | 10.04 | 6.28     |
| LSTM    | 0.9503 | 5.92 | 7.71  | 4.54     |
| GRU     | 0.9378 | 6.46 | 8.75  | 5.19     |

## Key Findings

- ▶ LSTM achieves the best overall performance with the highest  $R^2$  and lowest prediction errors.
- ▶ GRU performs better than XGBoost, while using a simpler recurrent architecture than LSTM.
- ▶ XGBoost remains interpretable and efficient, but neural networks capture nonlinear patterns more effectively.

# Model Configuration and Training Setup of GRU

## Deep Learning Training Setup

| Setting          | GRU  |
|------------------|------|
| Epochs           | 500  |
| Batch Size       | 16   |
| Validation Split | 0.20 |
| Early Stopping   | Yes  |
| Optimizer        | Adam |
| Loss Function    | MSE  |

Thank You

Questions?

Oshi Johri  
IIT Kanpur